

ATTENUATION OF A PERIODIC ACOUSTIC METAMATERIAL FOR MULTI-TONAL NOISE ATTENUATION

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A periodic Helmholtz resonator for low-frequency noise at the finite element method simulation performance of porous Helmholtz resonators in the cavity and multiple are obtained. Abbad et al. [3] cavity Helmholtz resonator and observed that the TL was inance with a degradation of acustic metamaterial based on containing a damping material ass is attached is investigated tached to the membrane is a oposed metamaterial presents ily one peak is obtained using monstrated that the resonant TL peaks can be controlled by of the added mass.

and analysis of amaterial
 sonator with extended neck il in the form of a membrane s with cylindrical shape is at- rane. The radius of the added kness by H. The resonator is a porous material. Figure 2 numerical geometry and the UC). ched solid mass are modeled the solid mechanics mod- while the air inside the neck is

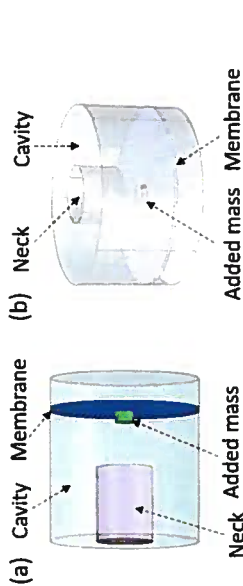


Figure 1: Helmholtz resonator with a membrane in the cavity (a) geometry, (b) numerical model.

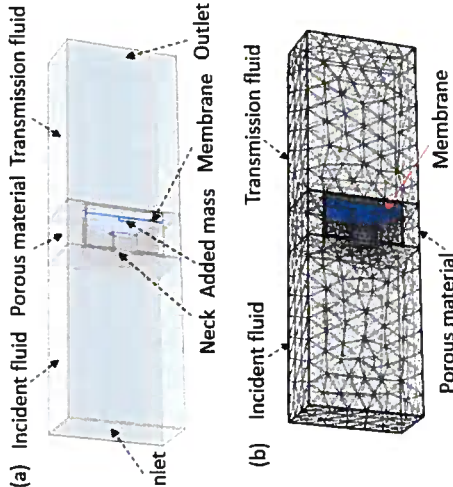


Figure 2: Cut-out view of the acoustic metamaterial (a) geometry of the PUC (b) mesh of the PUC.
 The membrane material is ethylene vinyl acetate rubber with a thickness of 1 mm, Young's modulus of 5 MPa, density of 660 kg/m³ and Poisson's ratio of 0.45. The radius of the neck is set to 15 mm and the diameter of the resonator cavity is 70 mm with a length of 40 mm, which is equal to the thickness of the porous material. Periodic boundary condi-

length of the neck is 15 mm and the membrane is located at 10 mm from the bottom inner wall of the resonator cavity. The TL with a conventional resonator shows one resonant peak of 43 dB at 692 Hz. When a membrane is integrated into the resonator cavity without added mass, the TL shows 2 peaks of 47 dB and 42 dB at 596 Hz and 1044 Hz respectively for membrane free boundary conditions and 5 peaks are observed with membrane fixed boundary conditions.

Figure 4 shows the effect of the added mass size on the TL. The length of the neck is set to 20 mm and the membrane with a thickness of 1 mm is located at 8 mm from the bottom inner wall of the cavity. The added mass is made of steel with a thickness of 2 mm and radius R varying from 1 mm to 6 mm.

With a radius R of 1 mm, the TL resonant frequencies in Fig.4 are 510 Hz, 612 Hz, 884 Hz and 1006 Hz where the TL peaks values are 29 dB, 38 dB, 30 dB and 27 dB respectively. With a radius of 4 mm, the resonant frequencies are 510 Hz, 646 Hz, 956 Hz and 1086 Hz. Apart from the first resonant frequency, the other three resonant frequencies with R= 4 mm have increased compared to R=1 mm. With R=6 mm, 3 resonance frequencies are observed which are respectively higher than the first three resonance frequencies with R=1 mm and R=4 mm. The TL resonance frequencies increase when the radius of the added mass increases. The resonance frequencies can then be controlled by adjusting the radius of the added mass.

Figure 5 illustrates the effect of the added mass material properties on the TL. The distance between the membrane and the bottom wall of the cavity is 8 mm. The length of the neck is 20 mm while R and H are set to 3 mm. The TL in Fig. 5 presents 6 resonant frequencies when the added mass is made of steel and 5 resonant frequencies are observed when the added mass is made of aluminum.

3 Conclusion

The TL of an acoustic metamaterial based on a periodic Helmholtz resonator was investigated using FEM. A membrane was integrated within the resonator cavity to which small solid masses were attached. It was demonstrated that the multiple TL resonant frequencies and peak values generated can be controlled by adjusting the parameters (size and material properties) of the added mass.

Acknowledgments

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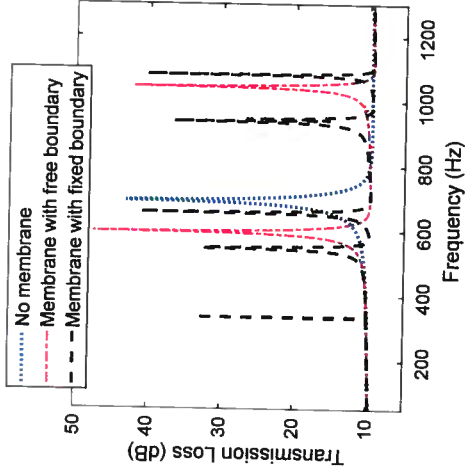


Figure 3: Comparison of the transmission loss.

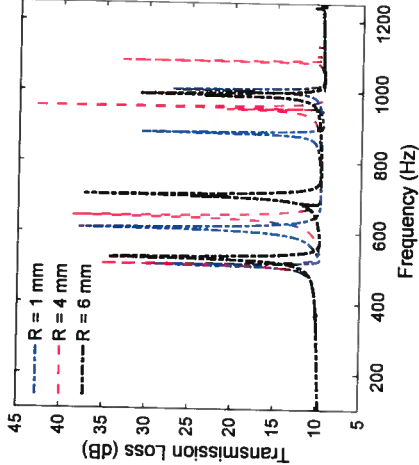


Figure 4: Effect of the added masses on the transmission loss.

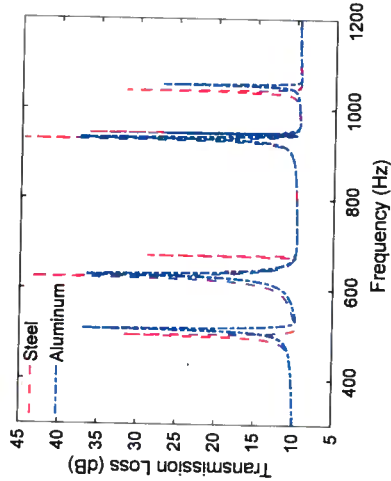


Figure 5: Effect of the added mass material on the transmission loss.